Chapter 1 Valuation Rings

Definition 1.0.0.1 Valuation rings:

- Let K be a field with $A, B \subseteq K$ local rings. We say that B **dominates** A if $A \subseteq B$ and $m_A = A \cap m_B$;
- Let A be a ring. We say A is a **valuation ring** if A is a local domain and if A is maximal for the relation of domination among local rings contained in the fraction field of A;
- Let A be a valuation ring with fraction field K. If $R \subseteq K$ is a subring of K, then we say A is **centered** on R if $R \subseteq A$.

◇ **Lemma 1.0.0.2** (Existence of Valuation Ring) Let K be a field and $A \subseteq K$ be a local ring. Then there exists a valuation ring with faction field K dominating A.

∘ **Proof** It is suffice to show: For collection of totally ordered collection of local subrings □

◇ **Lemma 1.0.0.3** Let A be a valuation ring, then A is a normal domain.

∘ **Proof** ... □

◇ **Lemma 1.0.0.4** Let A be a valuation ring with maximal ideal m and fraction field K, and $x \in K$. Then, either $x \in A$ or $x^{-1} \in A$ or both.

∘ **Proof** ... □

◇ **Lemma 1.0.0.5** ...

∘ **Proof** ... □

1.1 Valuation

◇ **Lemma 1.1.0.1** Let A be a valuation ring with field of fraction K. The set

...

∘ **Proof** ... □

Definition 1.1.0.2 Let A be a valuation ring:

- The totally ordered abelian group $(\Gamma, \geq)$ above is called **the value group** of valuation ring A;
- Map $v : A - \{0\} \to \Gamma$ and $v : K^* \to \Gamma$ is called valuation associated to A;

- The valuation ring A is called **discrete valuation ring** if $\Gamma \cong \mathbb{Z}$.

◇ **Lemma 1.1.0.3** ...

∘ **Proof** ... □

Chapter 2 Valuation Theory

Chapter 3 Ultrametric Space